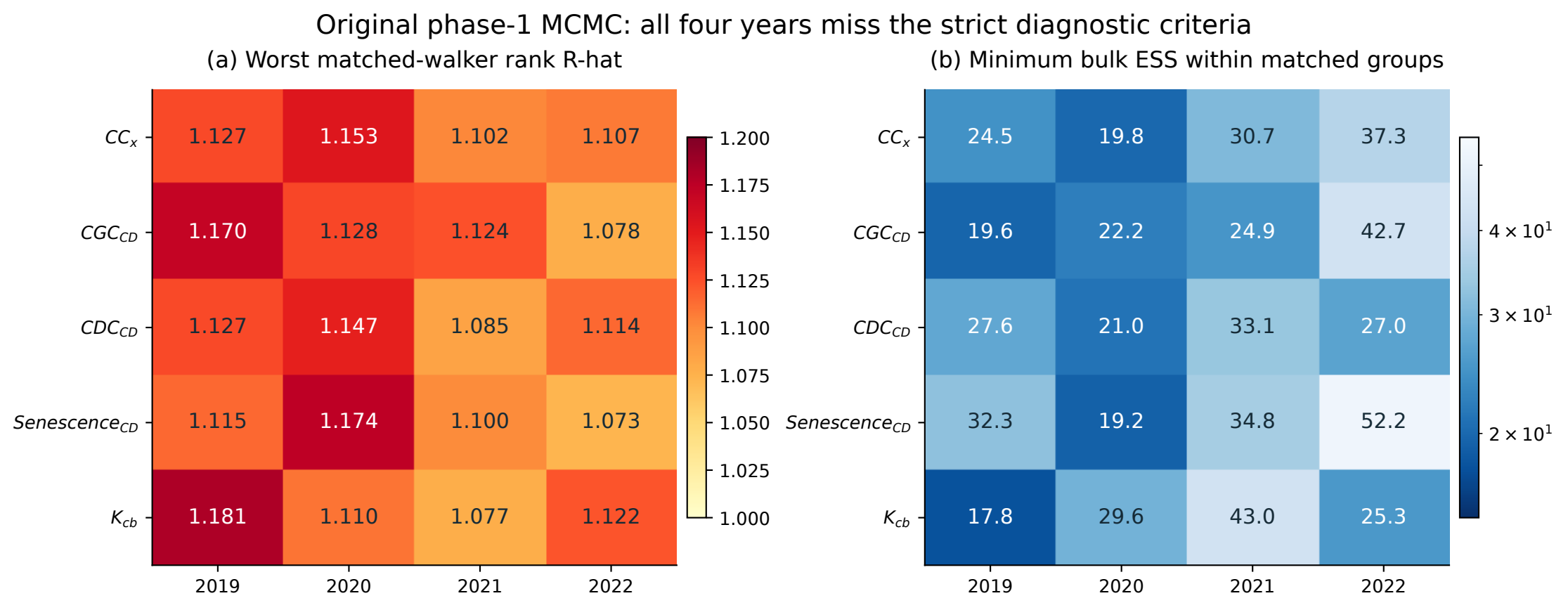
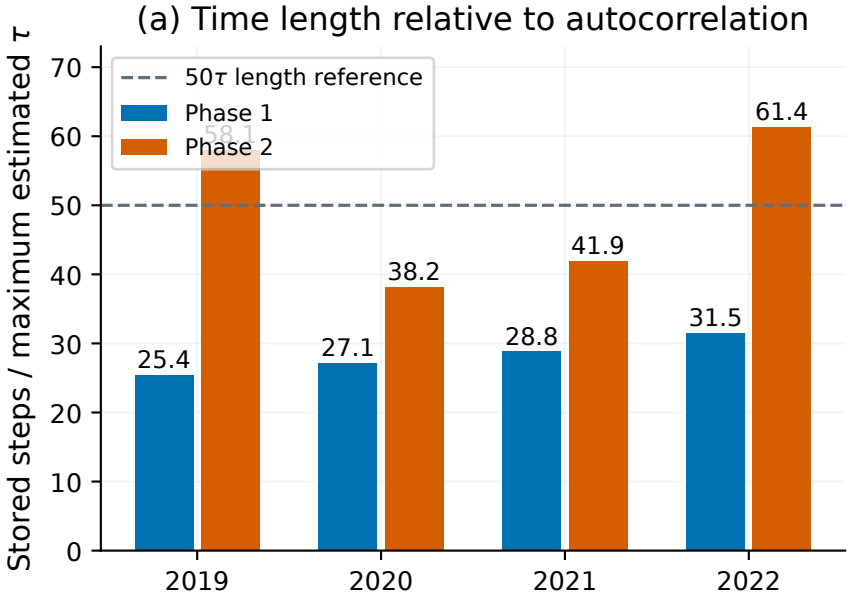
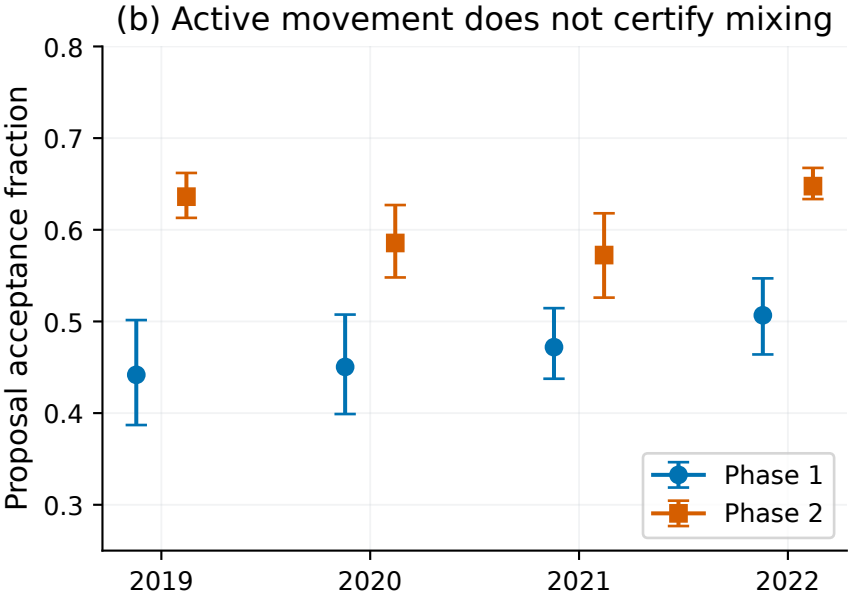

Original phase-1 MCMC: all four years miss the strict diagnostic criteria
(a) Worst matched-walker rank R-hat
(b) Minimum bulk ESS within matched groups
CC_x
CGC_{CD}
CDC_{CD}
$Senescence_{CD}$
K_{cb}
1.127 1.153 1.102 1.107
1.170 1.128 1.124 1.078
1.127 1.147 1.085 1.114
1.115 1.174 1.100 1.073
1.181 1.110 1.077 1.122
1.200 1.175 1.150 1.125 1.100 1.075 1.050 1.025 1.000
24.5 19.8 30.7 37.3
19.6 22.2 24.9 42.7
27.6 21.0 33.1 27.0
32.3 19.2 34.8 52.2
17.8 29.6 43.0 25.3
4×10^1
3×10^1
2×10^1
2019 2020 2021 2022
2019 2020 2021 2022

Original MCMC length and acceptance





All checkpoints contain 2000 steps per walker. The 50-tau rule is a length heuristic, not a convergence proof.

Phase 1: independent-ensemble trajectories

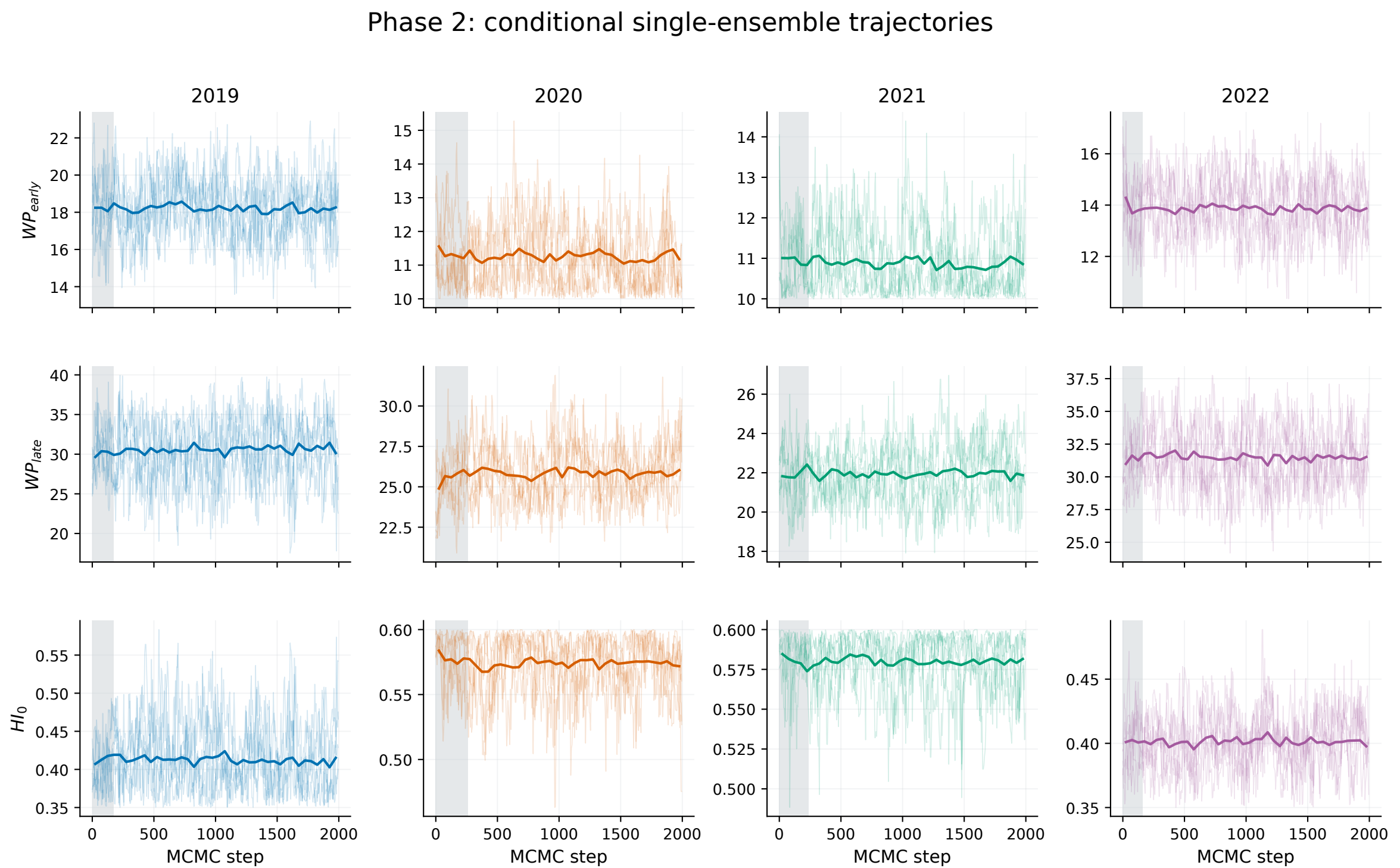
Ensemble 1 · Ensemble 2 · Ensemble 3 · Ensemble 4

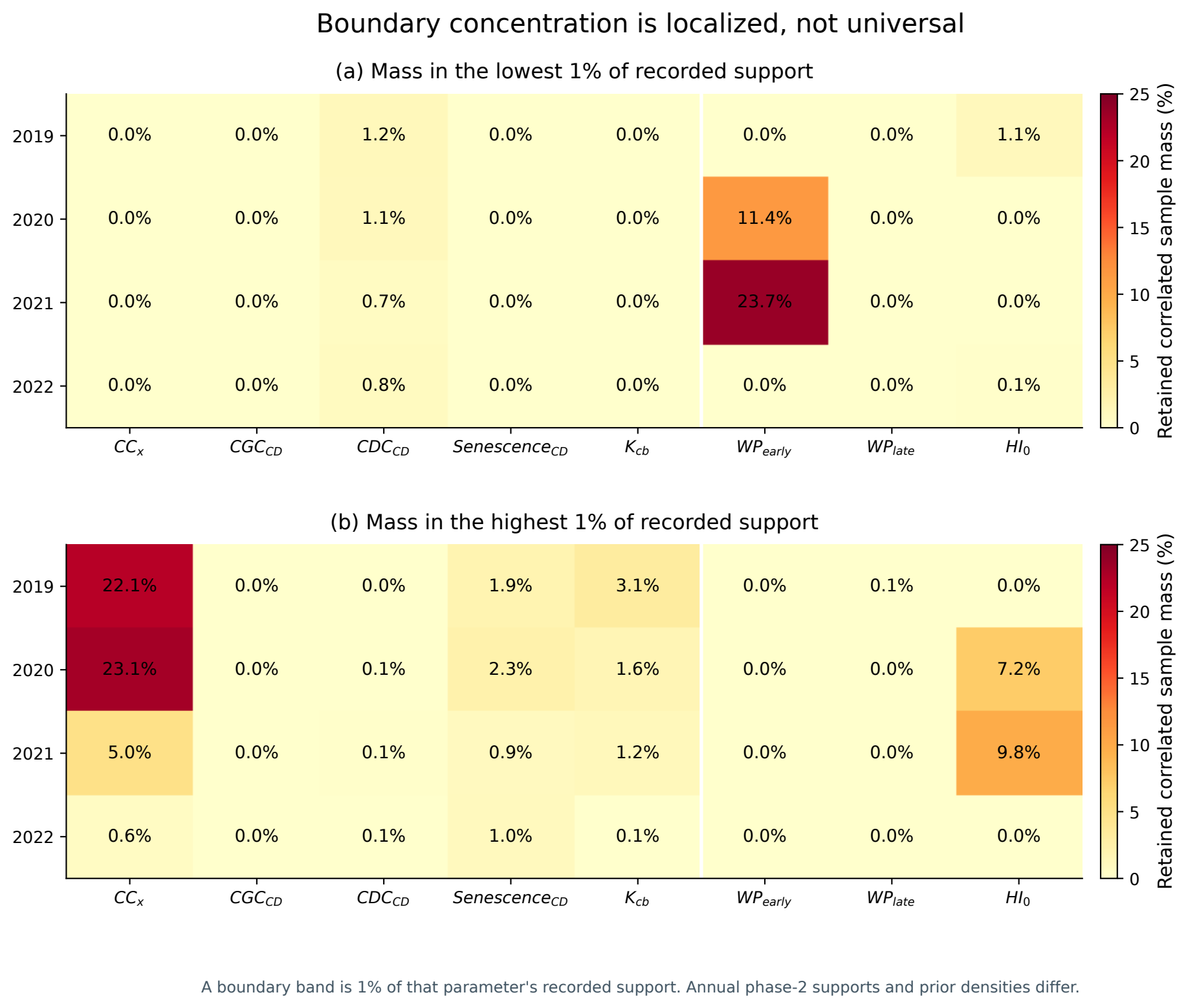
2019 · 2020 · 2021 · 2022

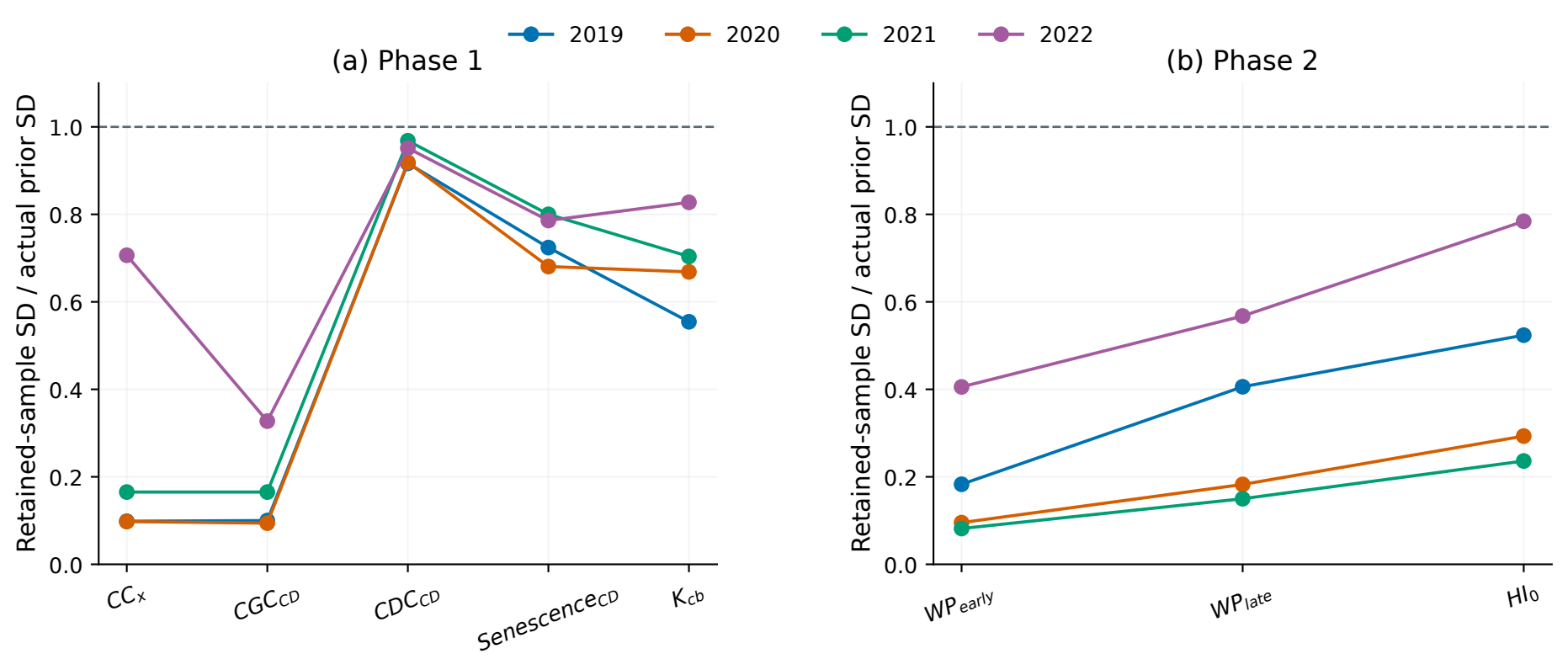
CC_x · CGC_{CD} · CDC_{CD} · $Senescence_{CD}$ · K_{cb}

MCMC step

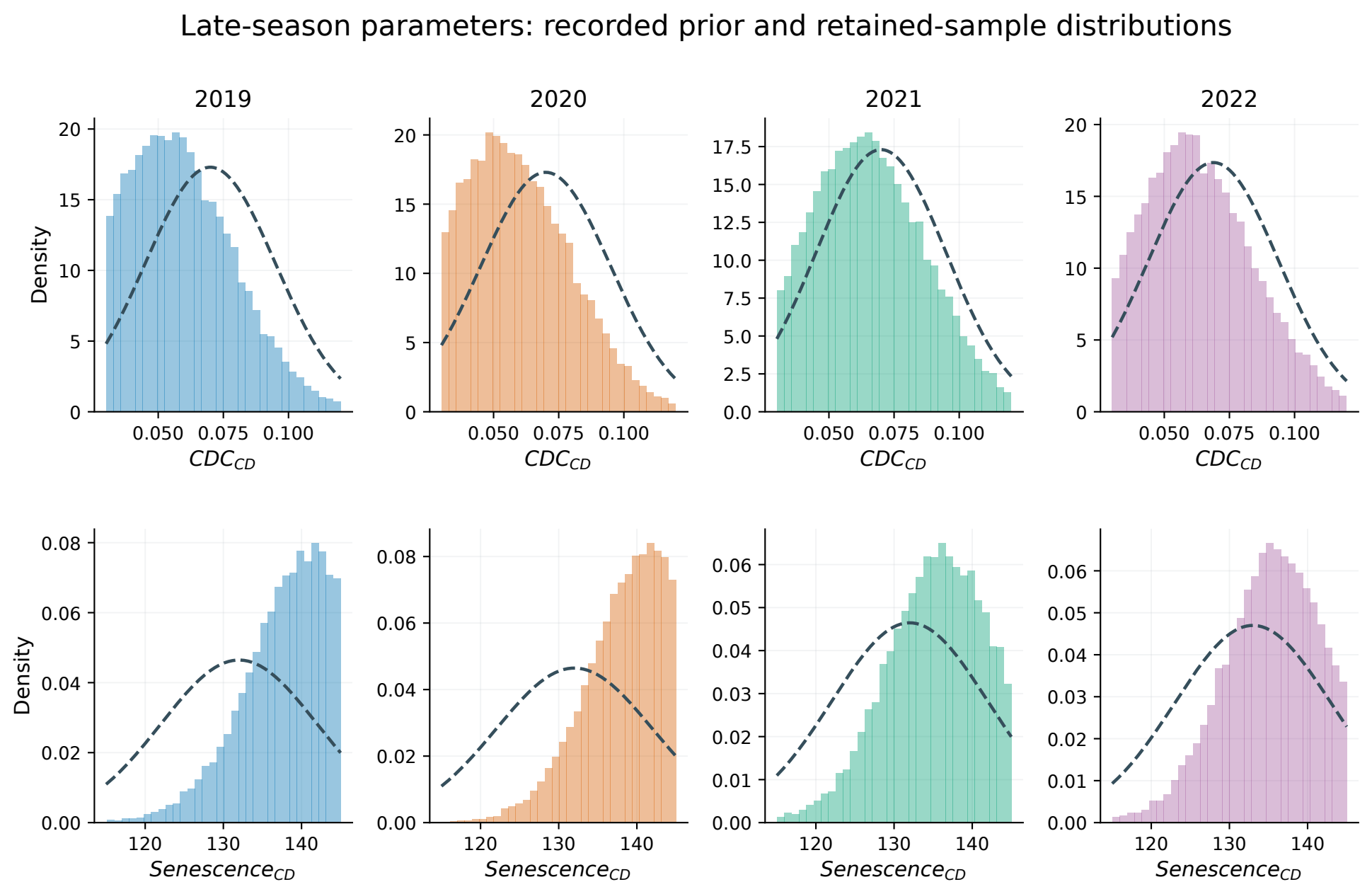
Lines are 50-step block means across 32 coupled walkers; gray areas are the original discarded warm-up.

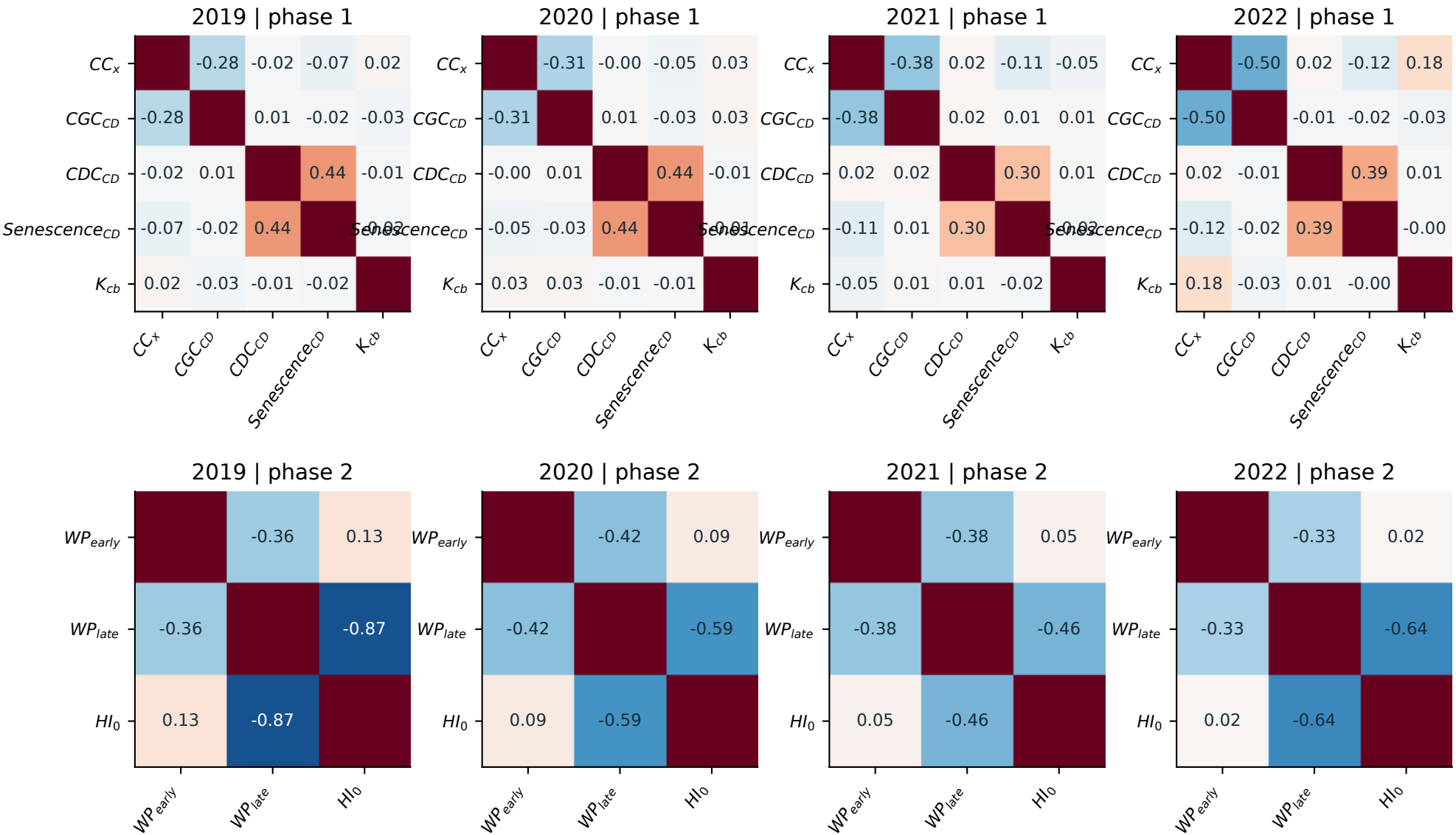

Phase 2: conditional single-ensemble trajectories
2019
2020
2021
2022
WP_{early}
WP_{late}
HI_0
MCMC step
Faint lines: six representative coupled walkers. Bold: 50-step block mean. Independent-ensemble R-hat is unavailable.


Boundary concentration is localized, not universal
(a) Mass in the lowest 1% of recorded support
2019
2020
2021
2022
0.0%
0.0%
1.2%
0.0%
0.0%
0.0%
0.0%
1.1%
0.0%
0.0%
1.1%
0.0%
0.0%
11.4%
0.0%
0.0%
0.0%
0.0%
0.7%
0.0%
0.0%
23.7%
0.0%
0.0%
0.0%
0.0%
0.8%
0.0%
0.0%
0.0%
0.0%
0.1%
CC_x
CGC_{CD}
CDC_{CD}
$Senescence_{CD}$
K_{cb}
WP_{early}
WP_{late}
HI_0
Retained correlated sample mass (%)
0
5
10
15
20
25
(b) Mass in the highest 1% of recorded support
2019
2020
2021
2022
22.1%
0.0%
0.0%
1.9%
3.1%
0.0%
0.1%
0.0%
23.1%
0.0%
0.1%
2.3%
1.6%
0.0%
0.0%
7.2%
5.0%
0.0%
0.1%
0.9%
1.2%
0.0%
0.0%
9.8%
0.6%
0.0%
0.1%
1.0%
0.1%
0.0%
0.0%
0.0%
CC_x
CGC_{CD}
CDC_{CD}
$Senescence_{CD}$
K_{cb}
WP_{early}
WP_{late}
HI_0
Retained correlated sample mass (%)
0
5
10
15
20
25
A boundary band is 1% of that parameter's recorded support. Annual phase-2 supports and prior densities differ.

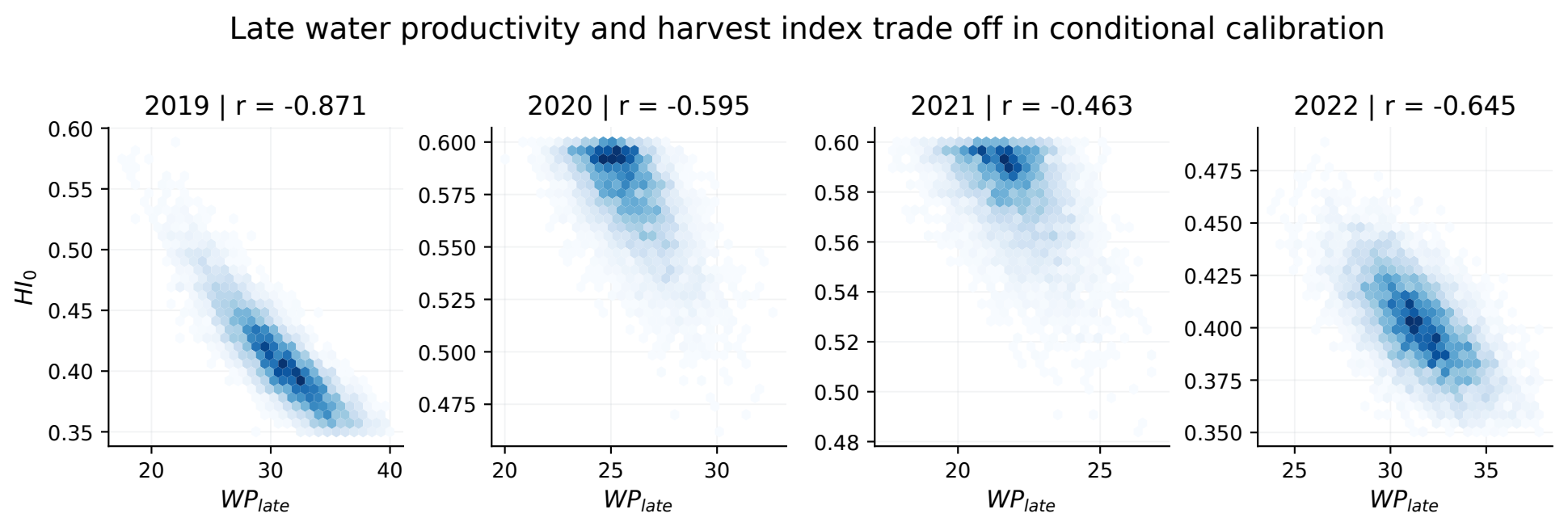


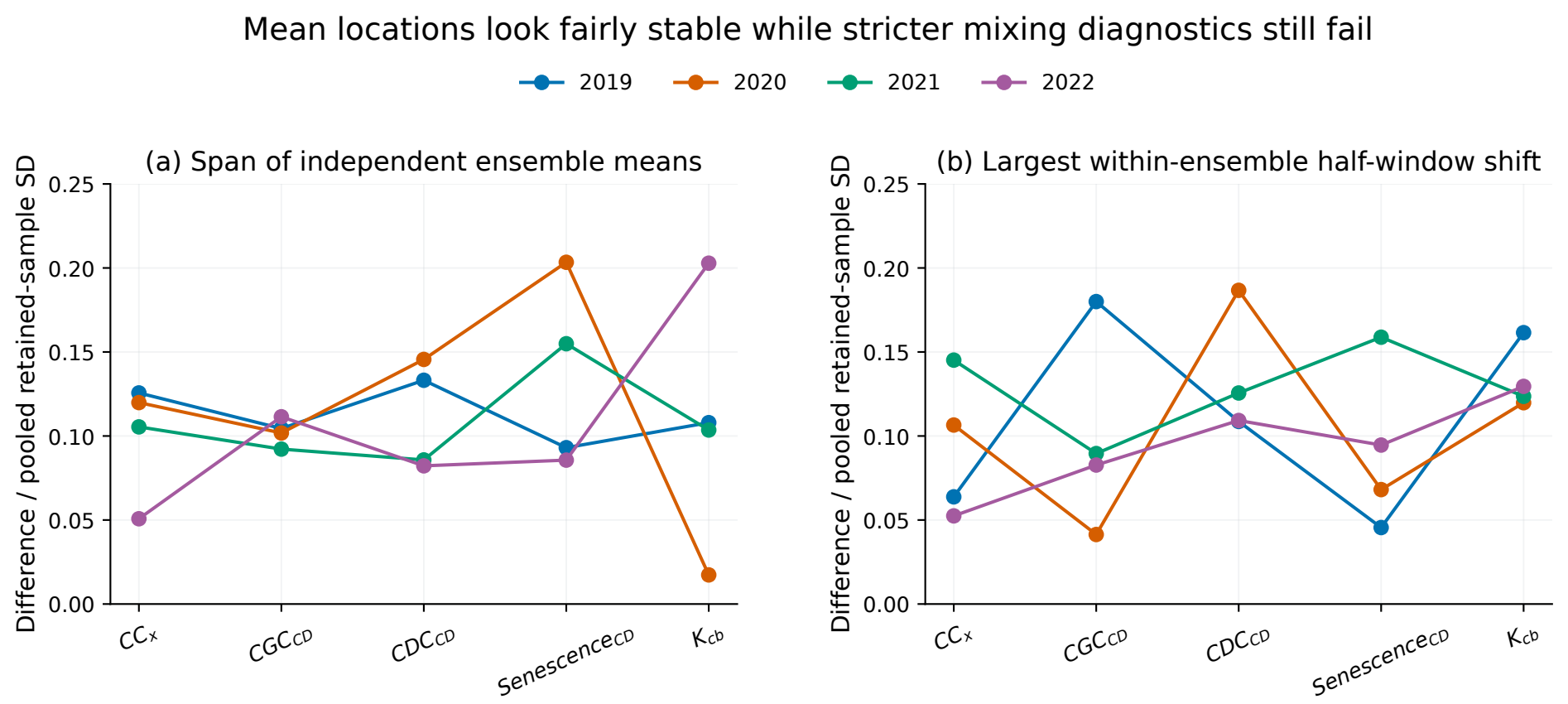
Ratios describe observed contraction only. Near-one values suggest limited learning under the specified prior; convergence remains uncertified.


Late-season parameters: recorded prior and retained-sample distributions
2019
2020
2021
2022
Density
CDC_{CD}
$Senescence_{CD}$
Dashed: actual truncated-normal prior. Filled histogram: phase-1 correlated sample distribution, deterministic plotting subsample.

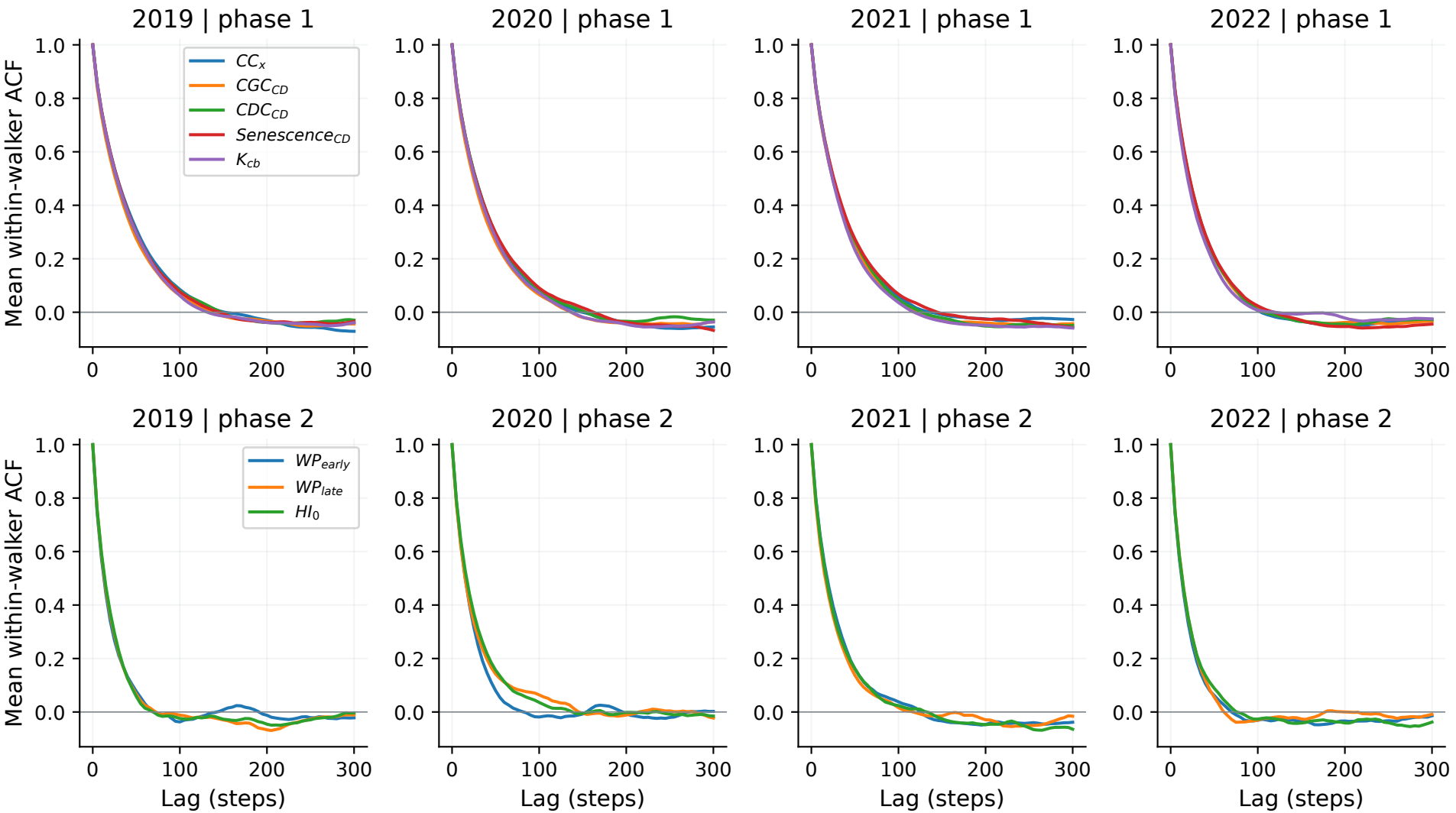


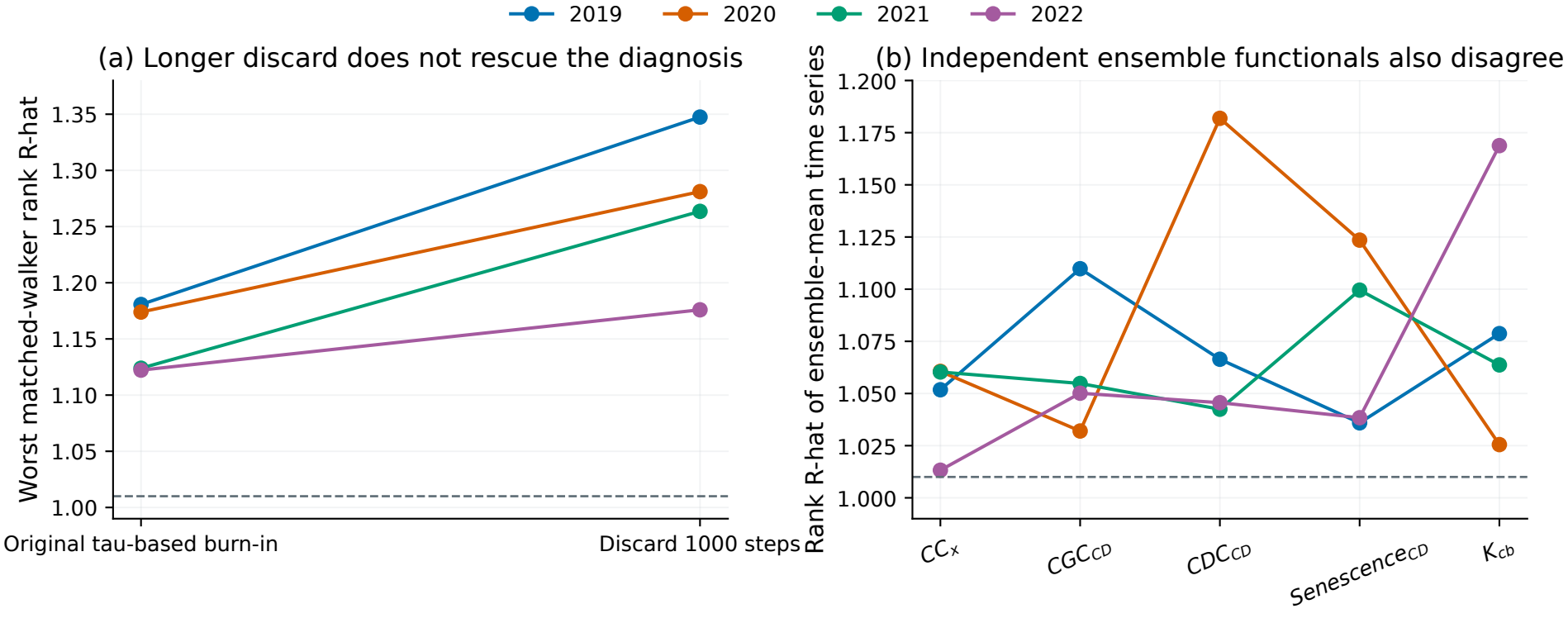
Pearson correlations from all retained rows; descriptive only. No cross-phase or cross-year joint posterior is constructed.


Late water productivity and harvest index trade off in conditional calibration
2019 | r = -0.871
2020 | r = -0.595
2021 | r = -0.463
2022 | r = -0.645
HI_0
WP_{late}
Density uses a deterministic sample subset; the correlations use all retained rows. Each year has one conditional ensemble.



No significance test is inferred from these correlated draws. Modest mean differences do not certify full-distribution convergence.


Autocorrelation persists despite frequent accepted proposals
2019 | phase 1
2020 | phase 1
2021 | phase 1
2022 | phase 1
2019 | phase 2
2020 | phase 2
2021 | phase 2
2022 | phase 2
Mean within-walker ACF
Lag (steps)
CC_x
CGC_{CD}
CDC_{CD}
$Senescence_{CD}$
K_{cb}
WP_{early}
WP_{late}
HI_0
ACFs are centered and normalized within each retained walker trajectory, then averaged; walker independence is not assumed.



Discard sensitivity is not a search for a passing window. Functional R-hat assesses that ensemble statistic, not pooled draw ESS.